The role of drought in plant diseases

Like other disease-causing agents, a plant pathogen (disease-causing agent) needs a supply of susceptible individuals to infect. Periods of drought slow the rate of plant reproduction and growth, thereby reducing the prevalence of disease.

Aridity, however, also weakens plants and makes them susceptible to pathogens that thrive in dry conditions. These include various forms of fungi that attack the leaves of grain crops, legumes, and fruits. These fungi are adapted to survive in a dormant state as hardened microscopic bodies in soil. They can exist for many years in dry soil, but when the soil becomes wet, the fungi must find a host within a few weeks or die. They do not necessarily kill their host. Recent research into chickpeas suggests that although infections from such fungi do increase during a dry spell, the mortality rate of the affected plants goes down during a drought.